

Secure Pharmacy Inventory and Prescription Management System with Medicine Stock Tracking

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INTRODUCTION

- Manages medicine inventory and patient prescriptions efficiently.
- Tracks medicine stock and expiry dates accurately.
- Reduces manual errors and paperwork.
- Stores data securely using **Structures, Arrays, Files, and Strings**.
- Improves the overall efficiency of pharmacy operations.
- The system reduces manual work and improves accuracy.
- It provides a secure, fast, and reliable solution for pharmacy operations.

PROBLEM STATEMENT

- Manual pharmacy management increases the chances of errors and delays.
- Growing demand for faster, secure, and accurate pharmacy services.
- Difficulty in tracking medicine stock and expiry dates in real time.
- Managing patient prescriptions and inventory records efficiently is challenging.
- Need for an automated system to improve accuracy, reduce manual effort, and enhance pharmacy operations.

ABSTRACT

- Securely manages pharmacy inventory and prescriptions.
- Tracks medicine stock and expiry dates.
- Uses **Structures, Arrays, Files, and Strings** in C.
- Reduces manual errors and improves efficiency.
- Provides secure and organized data management.

OBJECTIVES

- Automate pharmacy inventory management.
- Manage patient prescriptions efficiently.
- Monitor medicine stock and expiry.
- Reduce manual errors.
- Improve data security and accuracy.

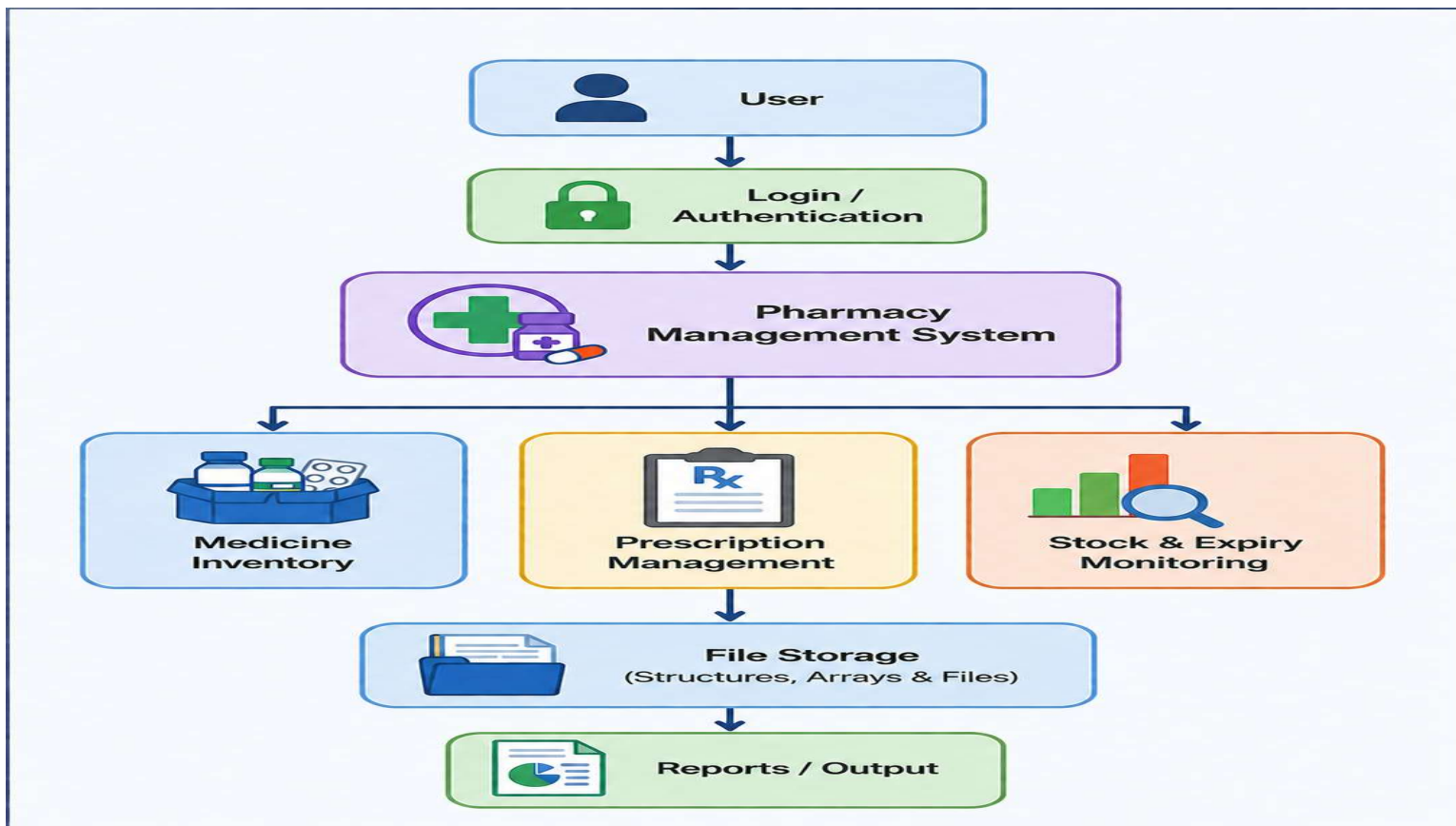
EXISTING SYSTEM VS PROPOSED SYSTEM

EXISTING SYSTEM	PROPOSED SYSTEM
Manual record maintenance	Automated pharmacy management
Paper-based prescriptions	Digital prescription management
Manual stock updates	Automatic stock tracking
No expiry monitoring	Expiry date alerts
More human errors	Improved accuracy and security
Time-consuming process	Fast and efficient operations

LITERATURE SURVEY

Author & Year	Research Focus	Limitation
Sinha et al. (2025)	Pharmaceutical Inventory Optimization	Focuses mainly on inventory optimization, not prescription management.
JPTCP Survey (2024)	Pharmacy Management System	Limited integration of stock tracking and expiry alerts.
IJSRET (2024)	Pharmacy Inventory & Prescription Management	Limited security and file-based implementation.
Our Proposed System	Secure Pharmacy Inventory & Prescription Management	Integrates inventory, prescriptions, stock monitoring, and expiry tracking in one system using C.

ARCHITECTURE OF THE PROJECT



PROJECT MODULES

Module	Description
Module 1: Medicine Inventory Management	Add, update, search, and delete medicine records. Maintain medicine inventory and availability.
Module 2: Prescription & Patient Medicine Management	Manage patient details, generate prescriptions, assign medicines, and maintain prescription records.
Module 3: Stock & Expiry Monitoring	Monitor stock levels, track expiry dates, identify low-stock medicines, and generate stock reports.

ALGORITHMS

Algorithm	Purpose
Linear Search	Search medicine records
CRUD Operations	Add, Update, Delete records
Stock Update	Update medicine quantity
Expiry Check	Detect expired medicines
File Handling	Store and retrieve data
Data Validation	Validate user inputs

CONCLUSION

- Successfully developed a secure pharmacy management system.
- Simplified inventory and prescription management.
- Improved medicine stock and expiry tracking.
- Reduced manual errors and processing time.
- Enhanced the efficiency and reliability of pharmacy operations.

ANY QUESTIONS....?



THANK YOU